3 Literature Review

With the continued rise of artificial intelligence, companies, large or small, must adapt to include the latest technologies. While larger companies generally have the financial capabilities to implement AI solutions they wish to have, small and medium-sized enterprises require more flexibility. In this literature review, the history of artificial intelligence and definitions of AI solutions will be discussed in detail. A closer look at how the technologies are implemented in companies in general is taken preceding a specific look at small and medium-sized enterprises and potential implementation steps with AI. The review further explains the requirements of applying artificial intelligence in SMEs and the role of data throughout. Introducing artificial intelligence solutions into businesses assists in streamlining processes and automating repetitive tasks while also expanding competitiveness.

3.1 Artificial Intelligence

Artificial Intelligence, or AI, has made large strides in recent years, with generative chatbots such as ChatGPT, Gemini, and Copilot being utilized in households, schools, and businesses. The theme is relatively new to larger audiences, given that initial legislation has been passed in many countries worldwide, regulating the use of AI tools. The beginnings, however, were much more limited when the subject was first coined in the middle of the 20th century. After English mathematician Alan Turing developed a coding machine named *The Bombe* for the British government to decipher the Enigma code used by Germany in the Second World War, the term Artificial Intelligence was officially coined by Marvin Minsky and John McCarthy in 1956 at the Dartmouth Summer Research Project on Artificial Intelligence (DSRPAI) (Haenlein & Kaplan, 2019). Since then, the evolution of AI has faced many ebbs and flows. Behind the first natural language processing tools and machine learning algorithms, which were created after the DSRPAI conference, the United States (U.S.) Congress criticized the amount of spending allocated for AI research (Haenlein & Kaplan, 2019). The denunciation brought an almost complete halt to progress in the field, not only in the U.S. but also in other countries such as the United Kingdom. A significant concern was that the technological capabilities of the Artificial Intelligence solutions being created during the 20th century were limited. From the 2010s, computers and machines with faster computing power have made it possible to work with a wide variety of Artificial Intelligence tools. The applications can be found almost everywhere nowadays, whether it be image recognition algorithms on Facebook, speech recognition algorithms with smart speakers, or machine learning algorithms throughout production at a company's plant (Haenlein & Kaplan, 2019). While AI has existed for more than half a century, there are still question marks as to what the future may look like with further progression of algorithms.

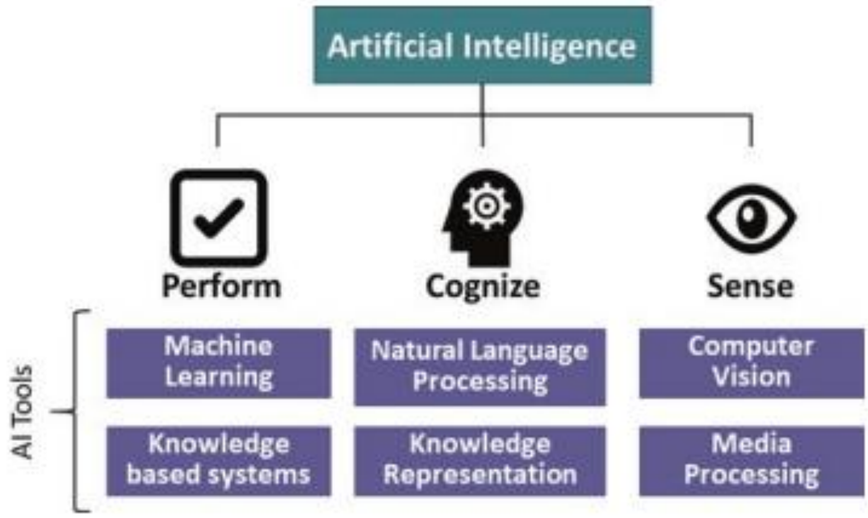

Figure 3: Definition of Artificial Intelligence (Akerkar, 2018)

'Reproduced with permission from Springer Nature'

With the state of Artificial Intelligence now, it is defined as the diverse range of technologies and tools that are combined to sense, cognize and perform with the capability of learning and adjusting over time (Akerkar, 2018). Another definition from Bharadiya et al. (2023) states, "The phrase artificial intelligence is a generic one that may be used to describe any variety of computer software that participates in tasks that are analogous to those performed by humans, such as learning, planning, and problem-solving." As seen in Figure 3, there are AI tools that fit under each category of AI ideologies. Computer Vision and Image Processing contain algorithms and methods to process and train data based on media and what is seen visually. Cognitively, natural language processing and knowledge representation are intertwined to increase contextual understanding of data and create reasoning and decisions through continued preparation. Under performance, machine learning and knowledge-based systems are combined to identify patterns or rules in datasets and thus improve understanding and explain why these patterns or rules are executed. Artificial intelligence offers higher scalability, longevity, and improvement opportunities, leading to increased productivity, a decrease in costs, and minimized human error (Akerkar, 2018). With the explosion of artificial intelligence use, businesses are jumping onto its ideas to take advantage of the benefits. While there have been concerns voiced about AI replacing humans, it should rather be used as a supporting hand. While struggling with basic tasks, artificial intelligence is able to process and analyze large datasets quicker than humans (Bharadiya et al., 2023). The technology is helpful across a wide range of business sectors, offering machine learning algorithms to process tedious and repetitive tasks, cybersecurity detection algorithms to prevent cyber-attacks, and automation processes to manage customers, products, or productivity, to name a few. While the origins of artificial intelligence may be traced back to the 1950s, the capabilities were limited until the 21st century, when advanced technologies allowed for faster computing times and AI-based algorithms to work with much larger amounts of data.

With the current state of Artificial Intelligence, governments and regulatory bodies need to act with more speed in creating or updating frameworks regarding the use and creation of AI tools. Advancements are being made at a much faster pace, and whether it is being used in positive or negative ways, AI is impacting the world we live in on a daily basis. In 2024, the European Union published the EU AI Act, the first legal framework for artificial intelligence for member countries. The legislation is written in a way to be applied across all sectors, address potential risks, define terms and requirements relating to AI systems and other frameworks related to AI technologies, and use cases that are prohibited (Evas, 2024). While some legislation is somewhat related to artificial intelligence, Germany does not have any legislation specifically referencing AI. The first two chapters of the EU AI Act have been in effect since February 2, 2025, while the remaining statutes are fully effective by August 2, 2026 (Lindert, 2025). Chapter I of the EU AI Act defines artificial intelligence and outlines the scope, while the second chapter details prohibited AI practices. According to the European Union's AI Act, artificial intelligence system is defined as „a machine-based system that is designed to operate with varying levels of autonomy and that may exhibit adaptiveness after deployment, and that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments" (Gaudszun et al., 2025). The European Parliament describes an AI system as software that creates its own decisions, learns and adapts once deployed, and uses data to create desired outputs based on the reasoning and purpose for the creation of the process. Germany adopts the definition from the EU AI Act, but many other countries have differing opinions when it comes to defining artificial intelligence. Canada has a similar definition, albeit less broad, states within the United States of America have individual descriptions, and several countries, such as China, Japan, and the United Kingdom, have no official depictions of the topic (Gaudszun et al., 2025). The European Union's legislation applies to businesses and people creating, deploying, or providing AI systems in the EU but is not specific to working sectors. Other chapters of the current legislation regarding artificial intelligence in most of Europe cover high-risk systems, transparency obligations, general-purpose models, innovation support, governance, post-market monitoring, information sharing and market surveillance, and penalties. As artificial intelligence continues to progress, countries must release and update legislation to adapt to its changes.

3.2 AI Solutions

Artificial intelligence comprises a variety of technologies that offer machines the ability to perform tasks originally designed for humans. Within this area, several AI solutions have been created since the foundation of artificial intelligence in the 1950s, contributing to its development in varying methods. Machine Learning (ML) is the first sub-area of artificial intelligence, which creates algorithms and statistical models that allow for computer systems to learn from data, identify models, and make decisions or predictions on what it has learned during training (Nicoletti, 2025). Machine learning was the first AI solution to be founded. Alongside artificial intelligence, machine learning was first suggested in Alan Turing's paper "Computing Machinery and Intelligence" before being officially coined by Arthur Lee Samuel in 1959 (Akerkar, 2018). There are three types of machine learning: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning is the process where the system learns a specific task based on labelled examples, whereas unsupervised learning involves unlabeled datasets, relying on the machine to organize the data on its own without obtaining any information ahead of the training and validation phases. Reinforcement learning is when the system learns through each workflow and receives positive or negative feedback to improve its future training (Nicoletti, 2025). Numerous algorithms under machine learning may aid systems in testing, training and validating data and training a model, including linear regression, k-nearest neighbor, decision trees, random forest, k-means, gradient boosting, and neural networks.

Deep Learning, or DL, is an important discipline of ML. Rajendra Akerkar (2018) defines deep learning as "a class of machine learning techniques that exploit many layers of non-linear information processing for supervised or unsupervised feature extraction and transformation, and for pattern analysis and classification." Less effort is needed for systems to recognize, classify and categorize patterns when using deep learning rather than machine learning. This also means that large amounts of data can be processed and managed. Autoencoders, deep belief networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs) are four deep learning models that have demonstrated effectiveness in learning hierarchical feature representations and solving data-driven tasks. Deep learning is the basis of generative artificial intelligence (GAI) as it authorizes systems to comprehend patterns from vast data sets to create materials replicating human creativity (Nicoletti, 2025). Machine learning mostly relies on regularly sized or smaller datasets, while deep learning is capable of using datasets of increased proportions.

Computer vision (CV) is one of the quickest changing fields in artificial intelligence, as it is experiencing growth and innovative solutions. Computer applications are able to interpret and understand images and videos by extracting, analyzing, and processing their visual data (Nicoletti, 2025). Automated guided vehicles (AGVs) use computer vision and image recognition for cars that have the capability of self-driving. Automotive companies' autopilot functions use cameras, sensors, and radar with CV to control the vehicle and be alerted to any potential issues or other vehicles or persons around the autonomous car. Facial recognition is another example of image processing and computer vision, being applied by police operations, airports, international border checks, payment services, and smartphone security features (i.e., face identification). Quality Assurance is another area where CV may be used more in the future. Manufacturers may be able to find defects and anomalies in products, processes, or machines during production, leading to earlier detection of issues being able to be quickly fixed (Nicoletti, 2025). The sooner defects are discovered by visual processing applications, the quality of products being delivered will increase, along with the satisfaction levels of said customers.

Natural language processing, or NLP, is best explained as a mediator in the communication between man and machine (Nicoletti, 2025). Language has developed into a normal form of interaction between people, no matter the complexity or structure. Natural language, however, is not as simple for computers, making it a core goal for artificial intelligence to create programs that are able to fully comprehend and generate human language. There are complications though, that make it difficult for programs using NLP to be created: words and sentences may have more than one meaning, and simple sentences may still have complexities, making it difficult for the system to understand on its own. The first of three natural language processing use cases is text analytics. Text analytics, also known as document processing, is an advanced speech recognition system. These NLP algorithms function by extracting useful information from any size of text (Akerkar, 2018). For example, an order request that a manufacturing company uses NLP to extract the company name, address, product code and name, and have it separated so it is easier to process through their ERP system. Sentiment analysis also extracts text but identifies if the parts of the text are positive, negative, or neutral. Attitude, emotions, and opinions can be monitored, helping businesses understand customer trends, product opinions, or brand and service impressions (Akerkar, 2018). Translation services have already existed but language translation through natural language processing provides more consistent and accurate translations (Nicoletti, 2025).

Large language models, also known by the acronym LLM, have the ability to understand and generate text, images, or other types of content just as a human would. There are now countless examples of LLMs, including OpenAI's ChatGPT, Google's Gemini, and Meta's LlaMA. By using "self-training" to continuously learn grammatical and semantic information while being used, the models may be used as chatbots or translators for a variety of use cases across all industries (Nicoletti, 2025). "Revolutionizing the way we work and communicate with each other", generative artificial intelligence

(GAI) includes techniques of generating new content in the form of text, images, videos, or audio (Feuerriegel et al., 2023). The user inputs a request, and the GAI algorithm will create content based on the inquiry. Generative AI is relatively new compared to other AI functionalities, meaning that there are still limitations: incorrect outputs, copyright concerns, and environmental impact complications. More on the issues surrounding artificial intelligence will be discussed in section 3.6. Agentic AI is similar to GAI but is specifically built to achieve specific goals rather than having a general use. Examples of agentic AI are autonomous vehicles and personal assistants (Nicoletti, 2025). In the use case for DOLL Fahrzeugbau, Agentic AI will be used with Microsoft's Copilot to build an agent for the onboarding process for the human resources (HR) department.

An example of an LLM, Microsoft's Copilot enables DOLL Fahrzeugbau GmbH to use the techniques of generative artificial intelligence through chatbots and agentic AI. Copilot was initially released in early 2023 as Bing Chat, using OpenAI's GPT-3 and GPT-4 technology through a partnership established in 2019 (Narayanaswamy, 2024). Within the Microsoft 365 Copilot application, there is a general chatbot that can answer questions or provide feedback. The specialty of the program is the ability to create agents specified to achieve certain goals. Microsoft offers pre-defined agents that are already semi-built to ask questions and retrieve responses, act as a coach for writing, brainstorming, or learning, and create and maintain goals. It is also possible to create an agent from scratch.

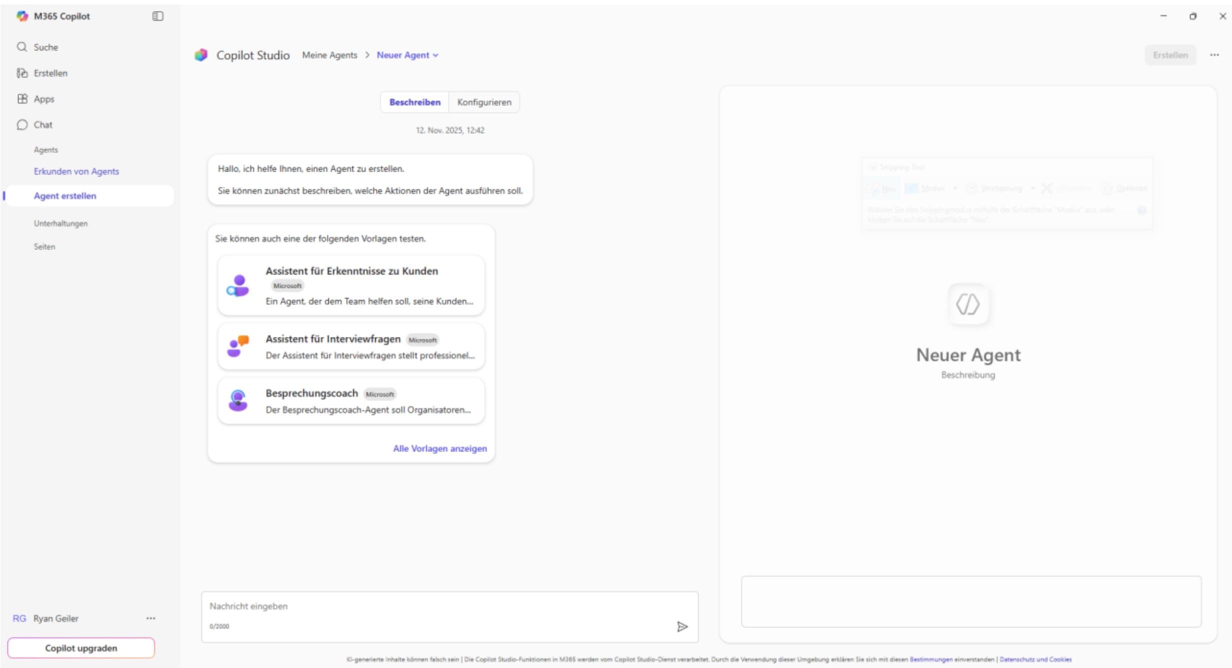

Figure 4: Microsoft Copilot's Agent Setup Page

In Copilot Studio, the builder mode displayed in Figure 4 asks questions related to the agent being created as answers and data are provided by the user wishing to manufacture the custom GPT

(Narayanaswamy, 2024). There is a manual method of creating agents, where the originator of the agent must provide a detailed overview of how the new chatbot should function, what sources it should use and learn from, and whether it has the ability to analyze data, perform calculations, create code snippets, or create visual aids. Agents may be connected to other Microsoft applications or some external applications that share their functionalities with Copilot. Microsoft's Copilot is one example of the progress large language models and artificial intelligence solutions have seen since their minute beginnings.

3.3 Implementation of AI

Artificial intelligence is increasingly being implemented in business to not replace human intelligence but to support it. Using the methods found in the previous section, AI-based processes and systems may be created and run to improve efficiency. There is a wide range of areas that AI may be used in business, and setting a scope is key in setting points and boundaries for the technology at any company. Whether used for assistance with operations, design, automation and human-machine interaction, or in quality assurance, the implementation of artificial intelligence is a step many companies have or are currently taking to further digitalize their portfolio.

In order to implement artificial intelligence, potential success factors should be conceived and considered if they are critical in the specific use case. In a research paper from the sixty-ninth volume of the International Journal of Information Management, nineteen factors impacting the implementation of artificial intelligence were identified, with some being IT infrastructure, integration complexity, low data quality, top management support, and selection of vendors (Merhi, 2023). Artificial intelligence systems are used interchangeably with other systems. If enterprises do not have sufficient existing hardware and systems to support the integration of AI, these must be upgraded in order for the implementation stage not to face any errors. With the complexity of AI systems coexisting with other systems, it is important to connect them properly so that data can be used, as this is what artificial intelligence uses to make more accurate decisions and hypotheses. If both of these factors do not succeed, the algorithms will not function properly. On an organizational level, support of the implementation first comes from higher management within a company. Due to the higher costs and strategic importance of AI, encouragement from management throughout the project is beneficial. Lastly, the idea of selecting vendors is based on picking AI solutions that may be integrated and operate without obstructions. If these factors, along with the other fourteen that Mohammad Merhi (2023) devised, are successful, the integration of artificial intelligence is a process worth considering.

In logistics and manufacturing, the implementation of AI solutions could be beneficial across several sectors. In planning, these applications may be used to help with predictive or smart maintenance, quality assurance, energy consumption forecasting, or supply chain management. Smart maintenance uses AI algorithms to predict when asset malfunctions occur to reduce unplanned downtime (Buchmeister et al., 2019), whereas predictive maintenance analyzes equipment sensor data to anticipate failures and plan maintenance schedules (Plathottam et al., 2023). This helps avoid most, if not all, material and financial losses due to disturbances or downtimes in manufacturing and logistics processes. In supply chain management, AI and machine learning are used to optimize supply chains and manage inventory levels and production planning using real-time data (Plathottam et al., 2023). Other primary functions that may see more automation include warehousing, for warehouse robots and recommendations on optimal locations for products and services, transportation, where automated guided vehicles (AGVs) and route optimization software for route planning are used, and process optimization, a basis for improving all aspects of manufacturing that function together (Nicoletti, 2025). With there being substantially more primary, secondary, and integration functions, we will focus on these in depth: procurement, quality assurance, and product design. Artificial intelligence is efficient when dealing with partner selection and handling documents. Algorithms are built to evaluate the performance of partners by analyzing existing or forecasting future lead times, costs, quality, and reliability factors (Nicoletti, 2025). By doing this, relationship management improves and insights into partners are easily readable. Quality assurance, with the assistance of automated tasks, will help improve the quality of products. Using image processing or natural language processing, visual defects, improper packaging, and labeling errors may be detected in real-time or notices about improper or potential issues from data in the ERP system may be given to those responsible (Nicoletti, 2025). This helps reduce the time and cost of manual inspections and provides better transparency within the manufacturing and logistics network. Finally, AI, paired with generative design, authorizes potential new products or processes to be created based on user-provided requirements (Plathottam et al., 2023). Already existing designs are used for training and used as the basis for the new designs, leading to improved customization and performance. While there are more functions and areas outside of manufacturing and logistics that would see the benefits of artificial intelligence-based algorithms, these consequences give an ideal basis for AI systems to be used for companies with a heavy focus on manufacturing and logistics.

Implementing artificial intelligence has a large impact on any company. Depending on the use cases the business has created in relation to AI, business processes will see alterations and further developments are possible. Machine learning and robotic process automation (RPA) are used to perform repetitive and tedious tasks, large amounts of data can be analyzed in order to provide faster insights, and customer experience can be personalized to exactly tailor their wants and needs (Bharadiya, 2023). Chatbots, virtual assistants, and voice recognition systems with the use of NLP can understand and interact with human language more proficiently, and HR processes can be enhanced with the introduction of assisting the candidate process for job applications, employee onboarding, and employee management. With the large amounts of documents and potentially a large number of applicants, creating AI-based algorithms to assist in these processes would increase efficiency, and time may be spent elsewhere. Artificial intelligence will continue to expand and improve over time, as the more data it is trained on, the more the system will understand what its tasks are. With this in mind, AI models will have to improve on understanding and generating human language as well as explaining their decisions. Robotics and autonomous systems may perform physical tasks, and any threats or disruptions may be detected and responded in order to improve cybersecurity (Bharadiya, 2023). While AI offers significant opportunities for organizations looking to implement its solutions, the critical success factors and functionalities must be thought of and reviewed before the actual implementation phase commences.

3.4 Small and Medium-Sized Enterprises

Businesses come in numerous shapes and sizes. Most, however, can be classified under the term small and medium-sized enterprises, or SMEs. In 2003, the European Commission standardized the definition for micro, small, and medium-sized enterprises, which countries under the jurisdiction of the European Union follow (Berisha & Pula, 2015). As of 2025, a new company size threshold has been added to the definition: small mid-cap enterprises. To be classified as an SME or SMC (small mid-cap), staff headcount and either turnover or balance sheet total must be taken into consideration. The threshold values are listed in Table 3.

Category	***Micro***	***Small***	***Medium-Sized***	***Small Mid-Cap***
Staff:	< 10	< 50	< 250	< 750
Turnover:	≤ €2 million	≤ €10 million	≤ €50 million	≤ €150 million
Balance Sheet:	≤ €2 million	≤ €10 million	≤ €43 million	≤ €129 million

Table 3: Company Size Thresholds (European Commission, 2025)

According to the German Statistisches Bundesamt - Destatis/StBA - (no date), the thresholds are the same except for the exception of SMCs. As small mid-caps are being introduced in 2025 by the

Directorate-General for Internal Market, Industry, Entrepreneurship and SMEs to the European Commission, individual countries of the EU (European Union) may not have implemented this definition yet. This is the case for Germany, along with others. In Germany, companies are labeled as large enterprises when they have more than 249 employees and more than 50 million euros in annual turnover (Statistisches Bundesamt - Destatis/StBA).

Category	Micro	Small	Medium-Sized
Employees:	< 10	> 10; ≤ 50	> 50; ≤ 300
Total Assets:	≤ $100,000	> $100,000; ≤ $3,000,000	> $3,000,000; ≤ $15,000,000
Total Annual Sales:	≤ $100,000	> $100,000; ≤ $3,000,000	> $3,000,000; ≤ $15,000,000

Table 4: SME Thresholds by World Bank (Berisha & Pula, 2015)

In Table 4, the definition of small and medium-sized enterprises from the World Bank is listed. When compared with Table 3, employee ranges are almost identical, but the measurements for the financial criterion are contrasting. The European Commission uses annual turnover and balance sheet total, whereas the World Bank uses total assets and total annual sales. While these are only two definitions of SMEs, each country has the freedom to define SMEs specific to the demands of the respective government agency. Ninety-nine percent of businesses in the European Union are classified as SMEs (European Commission). Due to the extensive range of the definition's aspects, there are relatively few enterprises that must register as a large business.

3.5 Requirements for AI Implementation in SMEs

With a clear definition of small and medium-sized enterprises and the foundation of artificial intelligence implementation presented, the requirements for SMEs when aiming to implement AI within their businesses are to be delineated. Small and medium-sized enterprises are important to the global economy, with there being a large number in Germany. Without digitalization, however, SMEs risk falling behind larger corporations. SMEs tend to struggle taking advantage of AI technologies, leading to difficulties when trying to transition to the use of AI-based solutions (Schönberger, 2023). Some benefits previously mentioned are then excluded, and the enterprise is unable to use the full potential of artificial intelligence. Introducing AI is increasing in importance as refusing any form of digitalization threatens long-term potential. In Marius Schönberger's research article in the Journal of Business Management (2023), it states, „the integration of digital technologies in SMEs is no longer an option, but an urgent need to remain competitive and adapt to the dynamic changes in the business world." From the 2022 Digital Economy and Society Index, published by the European Commission, of the twenty-seven member states of the European Union, only eight percent of SMEs are said to be using artificial intelligence in some form. Germany is slightly above average at eleven percent of SMEs in the country already having implemented AI in some capacity. With the crucial role small and medium-sized enterprises play in the global and local economies, and the benefits of artificial intelligence, including increased efficiency, productivity, and decision-making (Schönberger, 2023), the implementation of AI is an important step for SMEs to consider making sooner rather than later.

When an SME is considering introducing AI solutions into its existing ecosystem, there are additional requirements to those previously mentioned for all companies in section 3.3. For technological requirements, small and medium-sized enterprises should have a capable digital infrastructure, flexible and easy-to-use systems, and data management. IT infrastructure, compatibility with existing systems, and a readiness plan are key initial requirements as AI implementation is directly linked to the entire infrastructure of a company. User-friendly AI tools allow SMEs to adopt AI through basic training, leading to simplistic yet principal solutions being used (Ayinaddis, 2025). Data quality is a demanding aspect, and oftentimes, SMEs do not have adequate collection and removal standards, with the process being very time-consuming for smaller businesses (Oldemeyer et al., 2024). This requirement is often difficult for SMEs, as they may lack having large, high-quality data. For organizational requirements, smaller companies lack expertise and do not have employees with competencies in the field of artificial intelligence. Financial and management support also play a role, with limited financial resources potentially impacting an SME's choice to invest in AI technologies, as well as the process requiring support from management. Since many SMEs do not have their own AI expertise, they often depend on external consultants. Available solutions focusing on requirements and implementing AI are limited and usually aimed towards large companies (Oldemeyer et al., 2024). There are some approaches available for SMEs that may better fit their structures, but they are not as common. With the European Union's AI Act and other data privacy and AI regulations enacted, these are legal requirements that any company, including small and medium-sized enterprises, must follow. If these requirements are observed, the implementation of artificial intelligence would be a doable task, but there are still future opportunities and directions for businesses to consider.

There are potentially new opportunities for SMEs once they implement artificial intelligence. An investigative report in the Management Review Quarterly (Oldemeyer et al., 2024) reveals, "… AI is mainly applicable in the domain of operations, leading to a perception of limited potential for AI development in other areas." Beyond operational functions, there could be use for AI in marketing, purchasing, customer service, and product development. There are low-budget AI applications available for smaller businesses that are concerned with AI's typically high investment value. The process of implementation is tedious and lengthy, meaning that the longer a company waits, the less competitive and innovative the business will be viewed by competitors and external users. When artificial intelligence is being introduced within a company, it is also important to promote knowledge and build trust in AI for the employees. User knowledge and skills of the artificial intelligence solutions will lead to improved contributions with the new software and help understand if the implementation phase is a success (Ghobakhloo et al., 2012). Ultimately, companies that decide to implement artificial intelligence will benefit from its efficiency and competitive edge. SMEs that invest money and time into affordable AI solutions will adapt long-term to today's world of increased digitalization.

3.6 Positives and Challenges of AI Application

Applying the knowledge of how artificial intelligence may be implemented in companies overall and specifically in manufacturing and logistical sectors, there are several challenges and limitations despite there also being promising benefits. Artificial intelligence provides personalized customer experiences by analyzing the customers' behavior patterns and preferences, future trends to make dynamic decisions, and detailed views into risks and susceptibilities. Sensitive data usage by AI is a concern, as well as integration complexities, costs, and ethics (Abu Samara et al., 2024). The ethics of AI algorithms is a major topic as they are meant to run fairly, but if biases are in any training data, the outcomes will be false or discriminatory. The application will support departments in any business, from HR to manufacturing to accounting. Although some aspects of work may be simplified, there is a risk of losing human empathy and emotion, job displacement, and receiving incorrect answers (Kirova & Boneva, 2024). With robots, automated guided vehicles, and AI algorithms seeing an increased use, some jobs are at risk of no longer being in high demand, while other jobs that are related to the implementation and upkeep of these systems are in increased demand. According to a study in the Journal of Economics and Management from the University of Economics in Katowice, there are four main barriers when implementing artificial intelligence in small and medium-sized enterprises: lack of trust, lack of knowledge, lack of infrastructure, and lack of resources. As previously stated, due to the potential of biases, some SMEs may still view AI as generally unreliable. For some companies, identifying AI solutions that would fit their business and understanding its capabilities and limitations is a strenuous process (Zavodna et al., 2024). SMEs also face challenges implementing artificial intelligence due to either not having the proper resources or equipment, as well as monetary and technical requirements. Due to the numerous benefits and challenges, going through with the undertaking is an arduous decision for businesses to make, especially small and medium-sized enterprises.

To have a better understanding of the thoughts that small and medium-sized enterprises may have with the implementation of artificial intelligence, two studies provide some insight into their concerns and the positives they view from it. Marius Schönberger surveyed 105 people in Germany, with most respondents identifying as medium-sized business owners (62.5%), while the rest are categorized as small (16.3%) and micro enterprises (13.5%) (Schönberger, 2023).

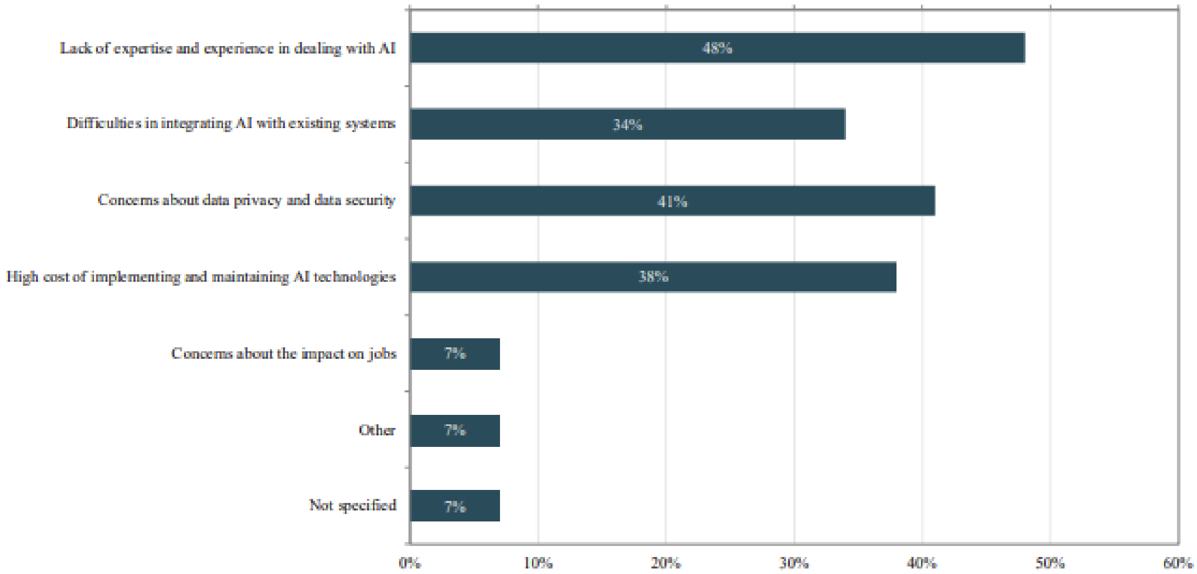

Figure 5: AI Implementation Challenges

Figure adapted from "Artificial Intelligence for Small and Medium-sized Enterprises: Identifying Key Application and Challenges" by Marius Schönberger, Journal of Business Management, 2023, licensed under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

Above, in Figure 5, the most prevalent challenges in the application of AI are listed with the percentage of respondents who answered for each response. Many SME owners or representatives agree that they lack expertise, would face difficulties in the integration process, have concerns about data privacy and security, and worry about the costs of implementing and maintaining the technologies. Although it was present as a concern in many of the references discussing the pros and cons of AI implementation, only 7% of those surveyed are concerned about the impact AI has on jobs. In Figure 6, the top three benefits of implementing AI technologies are as follows: improving business processes (59%), increasing efficiency and productivity (45%), and improving decision making (24%). More than twenty percent see it as an opportunity to develop new products or services, but this number may be influenced due to the areas of business the surveyors are in. Fourteen percent of the respondents believe cost reduction is a valid benefit as AI can reduce labor costs, streamline processes, and use resources more efficiently, but this is not seen by many in the survey as something that can outweigh the actual costs of implementing AI (Schönberger, 2023).

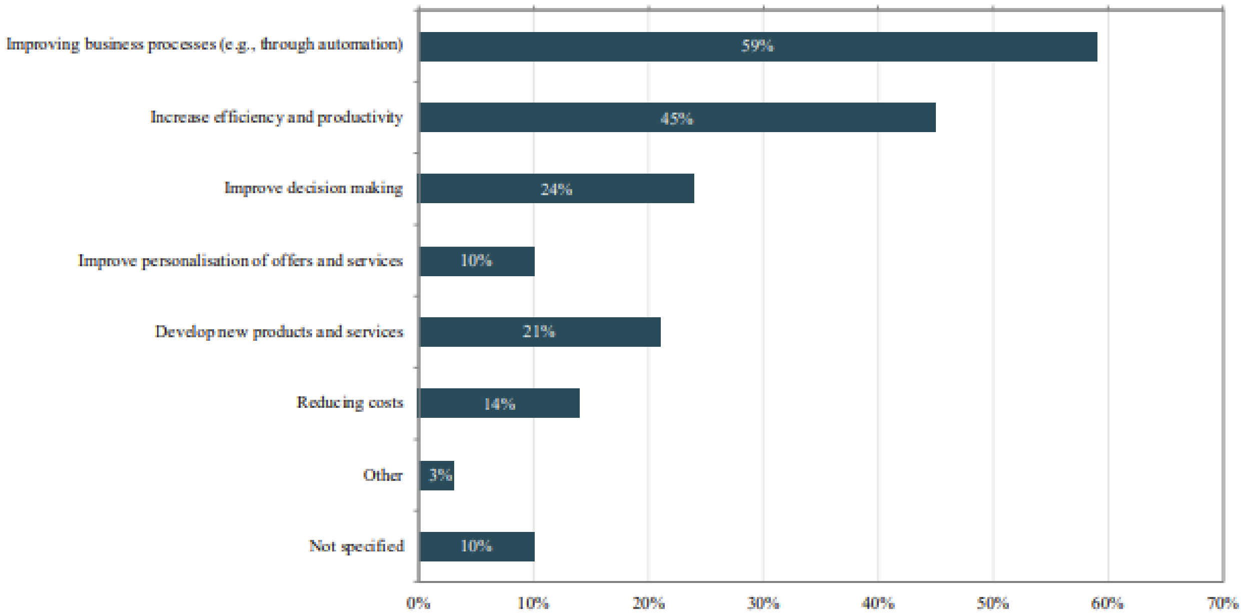

Figure 6: AI Implementation Benefits

Figure adapted from "Artificial Intelligence for Small and Medium-sized Enterprises: Identifying Key Application and Challenges" by Marius Schönberger, Journal of Business Management, 2023, licensed under CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

40 AI-experts and 98 representatives of SMEs from Germany participated in another survey to compare opinions on the benefits and potential complications of implementing AI. Figure 7 shows the chances of using artificial intelligence for various reasons in small and medium-sized enterprises between the experts and the SME representatives. The largest difference between the two is that SMEs believe it will aid in reducing staffing costs. This is also a complete difference from the previous study, although this study includes many more businesses in production and logistics, whereas the previous study had many respondents in management and IT. The representatives see the potential benefits of AI mostly in optimizing processes and innovating products and services. The AI experts agree but find that it is also key in developing new business models and increasing the quality of work.

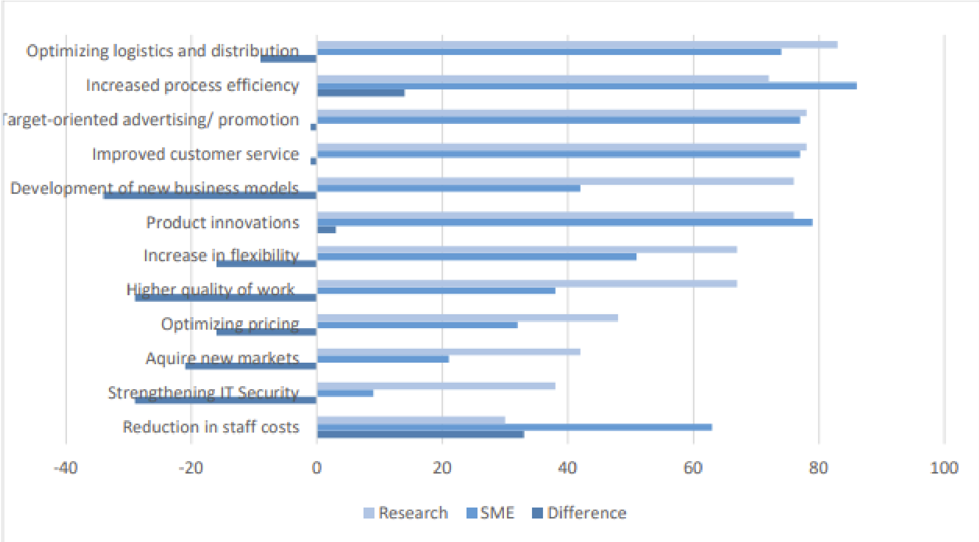

Figure 7: Chances of AI Use in Small and Medium-Sized Enterprises (Szedlak et al., 2021)

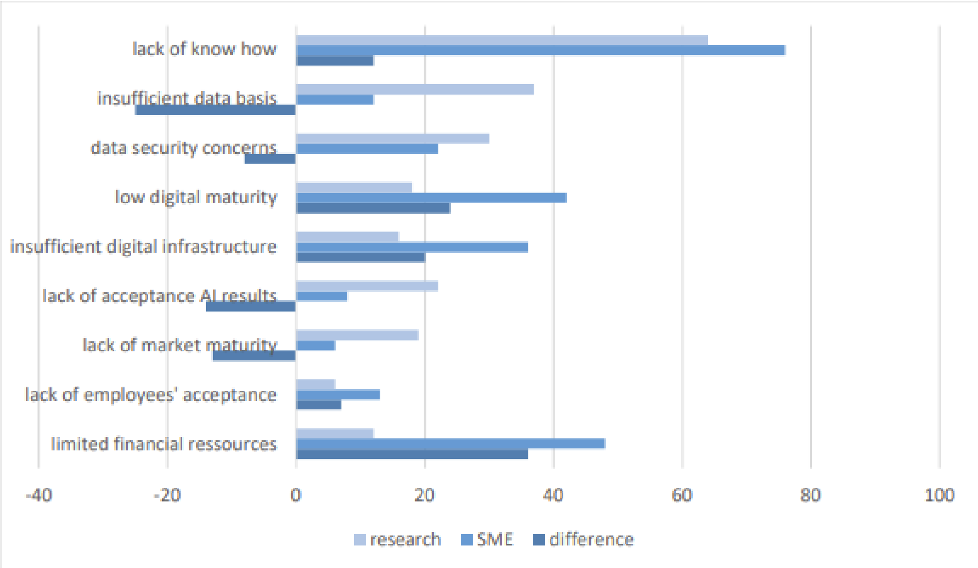

Figure 8: Barriers of AI Use in Small and Medium-Sized Enterprises (Szedlak et al., 2021)

In Figure 8, the importance of barriers to the implementation of AI, of the opinions of the AI-experts and the SME representatives surveyed, is shown. Both respondents agree that the lack of knowledge in knowing the steps in implementing artificial intelligence is the largest deterrence. SME representatives see issues with digital maturity and infrastructure as core barriers, while researchers lean more towards issues such as insufficient data basis, data security concerns, and trustworthiness of the solutions (Szedlak et al., 2021). While there are differences in what the core benefits and challenges are in implementing artificial intelligence, not only between researchers and SME representatives but also between the two surveys, we have a better outlook at the reasons why companies may choose to go through with the implementation of artificial intelligence or not.

3.7 Summary

This literature review has covered the foundation, development, and application of artificial intelligence in general, as well as in small and medium-sized enterprises. With a focus on SMEs, the review offered requirements and conditions relevant to smaller businesses. There are numerous concepts related to AI, such as machine learning, deep learning, computer vision, and generative AI, to name a few. These solutions are then to be implemented and used within companies, depending on their needs and use cases. Small and medium-sized enterprises were defined as they vary in size and value depending on where the business is located. The requisites small businesses need to have in order to implement AI solutions were listed, and ultimately, the benefits and drawbacks of adopting AI were considered. While AI has the ability to improve efficiency, decision-making, and competitiveness for SMEs, it still faces opposition with data quality concerns, trust issues, and high cost, which should be discussed before AI is adopted by small and medium-sized enterprises.